

NAAN MUDHALVAN PROJECT REPORT

**PROJECT TITILE : MICROGOALS – SECURE GOAL
MANAGEMENT BACKEND API**

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Project Overview

MicroGoals is a secure backend RESTful API developed using Node.js, Express.js, and MongoDB. It enables users to register, log in, and manage their personal goals through CRUD operations. The system uses JWT authentication and middleware to ensure data security and user-specific access control.

The system includes:

- User Registration & Login
- JWT-based Authentication
- CRUD Operations for Goals
- Protected Routes
- Centralized Error Handling

The main objective is to provide a secure and scalable backend system for digital goal management.

Project Instruction

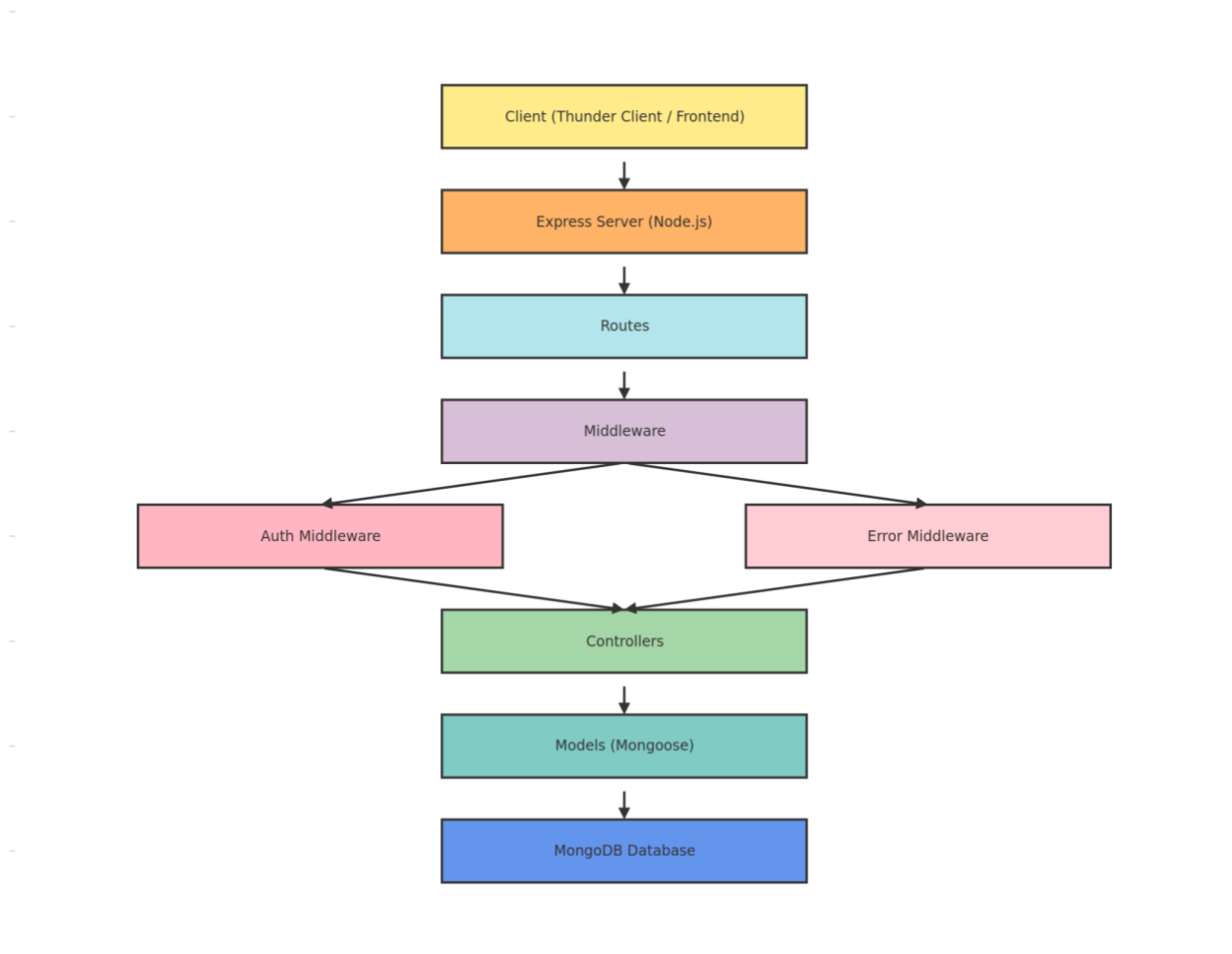
- The project follows REST API principles.
 - Authentication is handled using JWT.
 - MongoDB is used as the database.
 - APIs are tested using Thunder Client.
 - The project follows MVC architecture.
-

Pre-requisites

Before running the project, the following tools are required:

- Node.js (v18 or higher)
- Express.js
- MongoDB (Local / Atlas)
- VS Code
- Thunder Client (Extension)
- Basic knowledge of JavaScript & REST APIs

Project Architecture Image (Text Representation)



PROJECT ARCHITECTURE

TECHNICAL ARCHITECTURE

- Runtime: Node.js
 - Framework: Express.js
 - Database: MongoDB
 - Authentication: JWT
 - Architecture Pattern: MVC
-

ER DIAGRAM

Entities:

1. User

- `_id`
- `name`
- `email`
- `password`

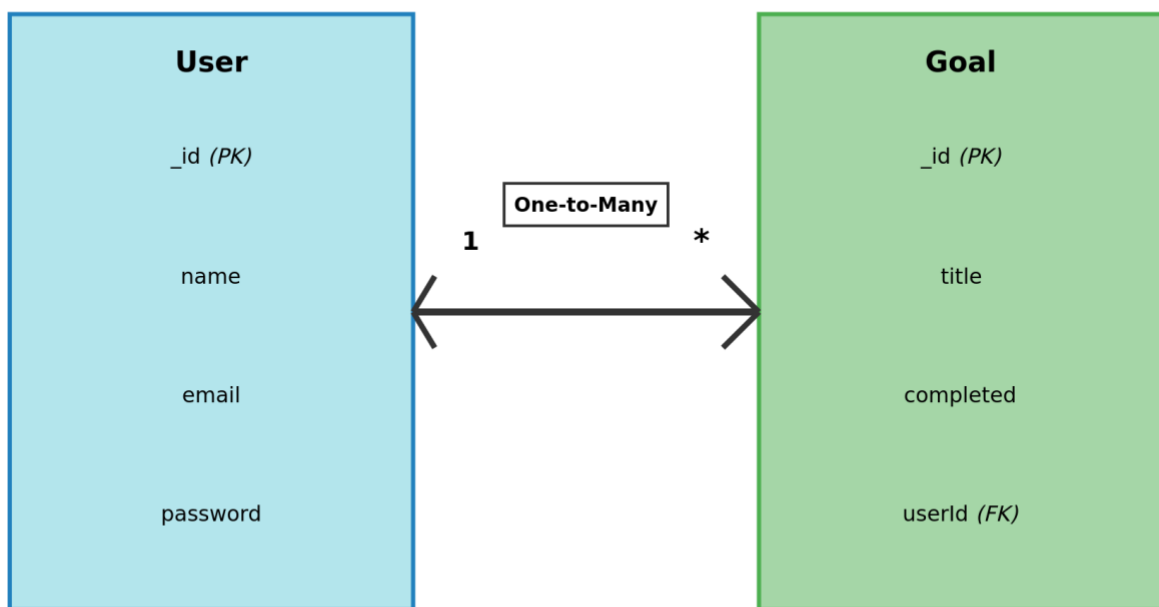
2. Goal

- `_id`
- `title`
- `completed`
- `userId` (Reference to User)

Relationship:

One User → Many Goals

User entities link to multiple Goal entities via foreign key



FEATURES

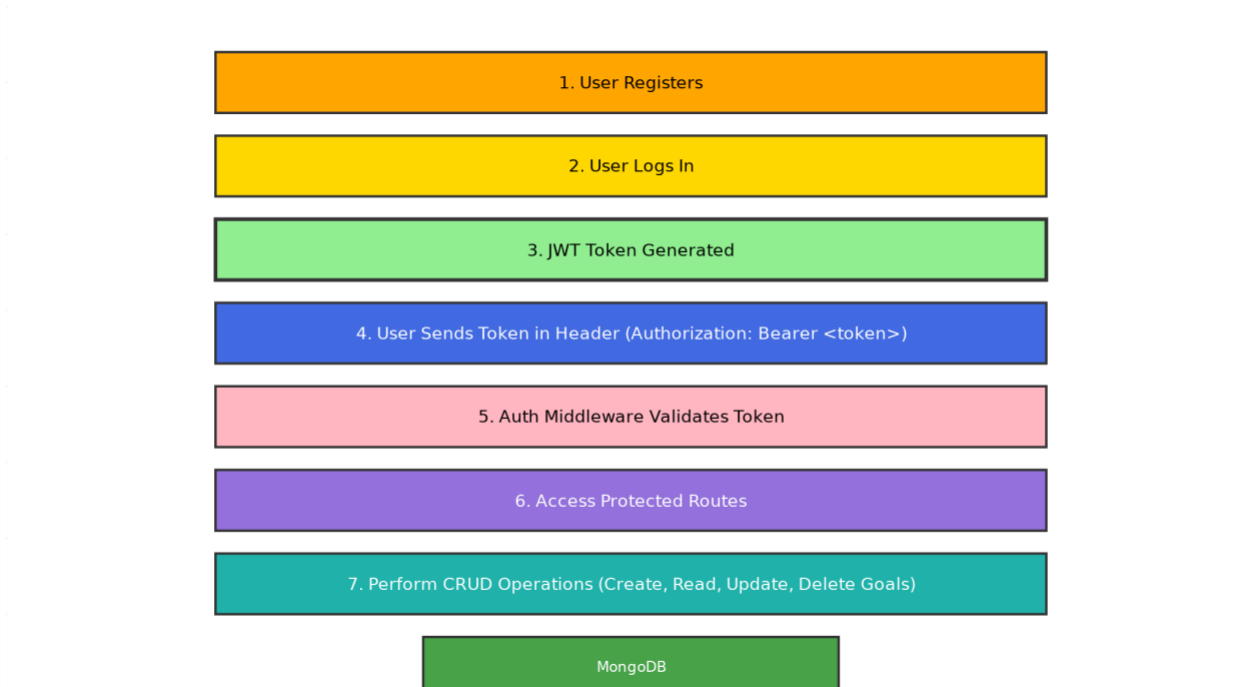
- User Registration
- Secure Login with JWT
- Create Goal
- Update Goal
- Delete Goal
- View User-specific Goals
- Protected Routes
- Centralized Error Handling

ROLES AND RESPONSIBILITIES

Role	Responsibility
User	Register, Login, Manage Goals
Developer	API Development & Testing

USERS FLOW

1. User registers
2. User logs in
3. JWT token generated
4. User sends token in header
5. Access protected routes
6. Perform CRUD operations



PROJECT SETUP AND CONFIGURATION

◆ Folder Structure

MicroGoals/

├── middleware/

| ├── authMiddleware.js

| └── errorMiddleware.js

├── Schema/

├── User.js

├── GoalSchema.js

├── models/

| ├── userModel.js

| └── goalModel.js

|

├── routes/

| ├── authRoutes.js

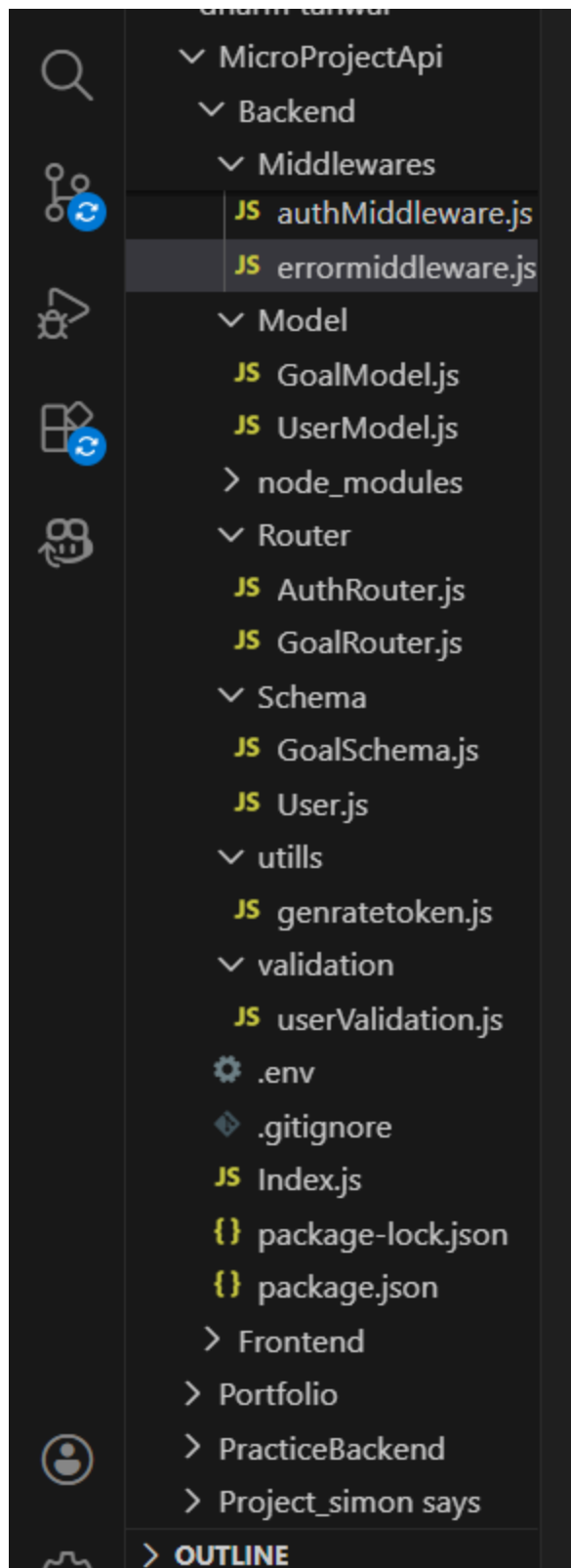
| └── goalRoutes.js

|

├── .env

├── Index.js

└── package.json



◆ Project Setup

1. npm init -y
2. Install dependencies:

```
npm install express mongoose jsonwebtoken bcryptjs dotenv
```

3. Start server:

```
node server.js
```

BACKEND DEVELOPMENT

- Created Express Server

An Express server was set up to handle API requests, connect routes, and enable JSON parsing for backend operations

```

JS Index.js X
1  require("dotenv").config();
2  const express = require("express");
3  const app = express();
4  const mongoose = require("mongoose");
5  const cors = require("cors");
6  app.use(cors());
7  const errormiddleware = require("../Middlewares/errormiddleware");
8  app.use(express.urlencoded({ extended: true }));
9  const bodyParser = require("body-parser");
10 app.use(express.json());
11 const PORT = process.env.PORT;
12 const Mongo_Url = process.env.MONGO_URL;
13 const authRoute = require("../Router/AuthRouter");
14 const GoalRouter = require("../Router/GoalRouter");
15
16 //mongoDB connection
17 mongoose
18   .connect(Mongo_Url)
19   .then(() => console.log("MongoDB Connected"))
20   .catch((err) => console.log(err));
21
22 //route
23 app.use("/", authRoute);
24 app.use("/", GoalRouter);
25
26 app.use(errormiddleware);
27 app.listen(PORT, () => {
28   console.log("Server running on port 8000");
29 });
30

```

- Connected MongoDB

For Connection with MongoDB:-

```

//mongoDB connection
mongoose
  .connect(Mongo_Url)
  .then(() => console.log("MongoDB Connected"))
  .catch((err) => console.log(err));

```

- Implemented Authentication APIs

User registration and login APIs were created with password hashing and JWT token generation for secure authentication.

```
JS Index.js JS AuthRouter.js X
1 const express = require("express");
2 const router = express.Router();
3 const bcrypt = require("bcryptjs");
4 const jwt = require("jsonwebtoken");
5 const userValidation = require("../validation/userValidation");
6 const userModel = require("../Model/UserModel");
7 const generateToken = require("../utils/genratetoken");
8 router.post("/signup", async (req, res, next) => {
9   try {
10     const { error } = userValidation.validate(req.body);
11     if (error) {
12       return res.status(400).json({
13         success: false,
14         message: "enter the data correctly",
15       });
16     }
17     const { name, email, password } = req.body;
18     const exist = await userModel.findOne({ email });
19     if (exist) {
20       return res.status(400).json({ message: "User already exists" });
21     }
22     const hashpassword = await bcrypt.hash(password, 10);
23     const user = new userModel({
24       name: name,
25       email: email,
26       password: hashpassword,
27     });
28     await user.save();
29     const token = generateToken(user._id);
30     res.status(200).json({
31       success: true,
32       message: "user is created",
33       token: token,
34       data: user,
35     });
36   } catch (err) {
37     return next(err);
38   }
39 });
40
41 router.post("/login", async (req, res, next) => {
42   try {
43     const { email, password } = req.body;
44     const user = await userModel.findOne({ email });
45
46     if (!user) {
47       const err = new Error("user not found");
48       err.status = 400;
49       return next(err);
50     }
51
52     const isMatch = await bcrypt.compare(password, user.password);
53     if (!isMatch) {
54       const err = new Error("Wrong Password");
55       err.status = 400;
56       return next(err);
57     }
58     const token = generateToken(user._id);
59     res.status(200).json({
60       success: true,
61       message: "login successfully",
62       token: token,
63       data: user,
64     });
65   } catch (error) {
66     return next(error);
67   }
68 });
```

- Implemented Goal CRUD APIs

Goal CRUD APIs were implemented to allow authenticated users to perform full Create, Read, Update, and Delete operations on their personal goals. These APIs follow RESTful architecture principles and are designed to ensure secure and structured data management.

The Create API enables users to add new goals to the database. Each goal is associated with a specific user using a user Id reference, ensuring user-specific data isolation.

The Read API allows authenticated users to retrieve only their own goals from the database. Data filtering is applied using the authenticated user's ID to prevent unauthorized data access.

The Update API enables users to modify existing goals. Proper validation checks ensure that only the owner of the goal can update it.

The Delete API allows users to permanently remove their goals from the system. Secure route protection ensures that users cannot delete goals belonging to others.

All CRUD routes are protected using Authentication Middleware, which verifies the JWT token before granting access. If the token is missing or invalid, the system returns a 401 Unauthorized error.

Additionally, Error Middleware is implemented to handle runtime and validation errors centrally. It captures exceptions from CRUD operations and returns consistent JSON responses with appropriate HTTP status codes such as 200, 201, 400, 401, and 404.

This structured implementation ensures secure data handling, proper authorization, and reliable API behavior across all goal management operations.

```
const express = require("express");

const router = express.Router();

const goalModel = require("../Model/GoalModel");

const authmiddleware = require("../Middlewares/authMiddleware");
```

```
router.post("/create", authmiddleware, async (req, res, next) => {

  try {

    const { title } = req.body;

    const goal = new goalModel({

      title: title,

      user: req.user,

    });

    await goal.save();

    res.status(200).json({

      success: true,

      data: goal,

    });

  } catch (err) {

    return next(err);

  }

});
```

```
router.get("/:id", authmiddleware, async (req, res, next) => {

  try {

    const id = req.params.id;

    const data = await goalModel.findOne({

      _id: id,

      user: req.user,
```

```
    });

    res.status(200).json({

      success: true,

      data: data,

    });

  } catch (error) {

    return next(error);

  }

});

router.put("/:id", authmiddleware, async (req, res, next) => {

  try {

    const goal = await goalModel.findOneAndUpdate(

      { _id: req.params.id, user: req.user },

      { completed: req.body.completed },

      { new: true },

    );

    res.status(200).json({

      success: true,

      message: "the goal is completed",

    });

  } catch (error) {

    return next(error);

  }

});
```

```

    }
  });

  router.delete("/:id", authmiddleware, async (req, res, next) => {

    try {

      await goalModel.findOneAndDelete({

        _id: req.params.id,

      });

      res.status(200).json({ message: "Goal deleted" });

    } catch (error) {

      next(error);

    }

  });

  module.exports = router;

```

- Added Auth Middleware
- Auth middleware was implemented to verify JWT tokens and secure protected routes from unauthorized access.


```
JS Index.js JS authMiddleware.js X
1  const jwt = require("jsonwebtoken");
2  require("dotenv").config();
3
4  const authmiddleware = (req, res, next) => {
5    try {
6      const header = req.headers.authorization;
7
8      if (!header) {
9        return res.status(401).json({ message: "No token" });
10     }
11
12     const token = header.split(" ")[1];
13
14     const decoded = jwt.verify(token, process.env.SECRET_KEY);
15
16     req.user = decoded.userId;
17
18     next();
19   } catch (error) {
20     return res.status(401).json({ message: "Invalid token" });
21   }
22 };
23 module.exports = authmiddleware;
24
```

- Added Error Middleware

An error middleware was added to handle application errors centrally and return proper status codes and messages.

JS *errormiddleware.js* ✕

```
1  const errormiddleware = (err, req, res, next) => {  
2    console.log(err);  
3    const status = err.status.code || 500;  
4    const message = err.message || "server error";  
5    res.status(status).json({  
6      success: false,  
7      message: message,  
8    });  
9  };  
10  
11  module.exports = errormiddleware;
```

DATABASE DEVELOPMENT

CONFIGURE MongoDB

- Use MongoDB Atlas or Local MongoDB
- Store connection string in `.env`

CREATE DATABASE CONNECTION

Index.js file:

- Use `mongoose.connect()`

```
//mongodb connection
mongoose
  .connect(Mongo_Url)
  .then(() => console.log("MongoDB Connected"))
  .catch((err) => console.log(err));
```

- Handle connection errors

CREATE SCHEMA AND MODELS

User Schema

The User Schema defines the user data structure with name, email (unique), and hashed password fields.

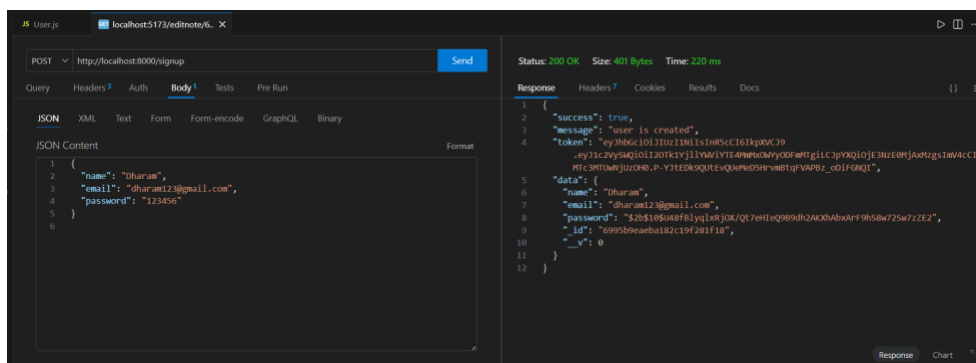
```
JS User.js X
1  const mongoose = require("mongoose");
2  const Schema = mongoose.Schema;
3  const userSchema = new Schema({
4    name: {
5      type: String,
6      required: true,
7    },
8    email: {
9      type: String,
10     unique: true,
11   },
12   password: {
13     type: String,
14     required: true,
15   },
16 });
17 module.exports=userSchema;
18
```

Goal Schema (with userId reference)

The Goal Schema stores goal details and includes a userId reference to associate each goal with a specific user.

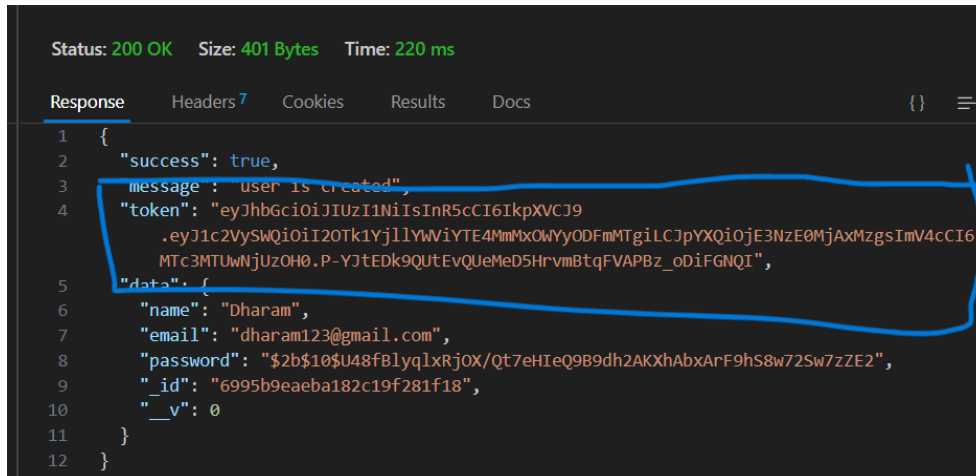
JS GoalSchema.js X

```
1  const mongoose = require("mongoose");
2  const Schema = mongoose.Schema;
3
4  const goalSchema = new Schema({
5    title: {
6      type: String,
7      required: true,
8      trim: true,
9    },
10
11    completed: {
12      type: Boolean,
13      default: false,
14    },
15
16    user: {
17      type: mongoose.Schema.Types.ObjectId,
18      ref: "User",
19      required: true,
20    },
21  });
22
23  module.exports=goalSchema;
24
```



- JWT Token Response

After successful login user the api generate the token and these token was sent with every request to every api

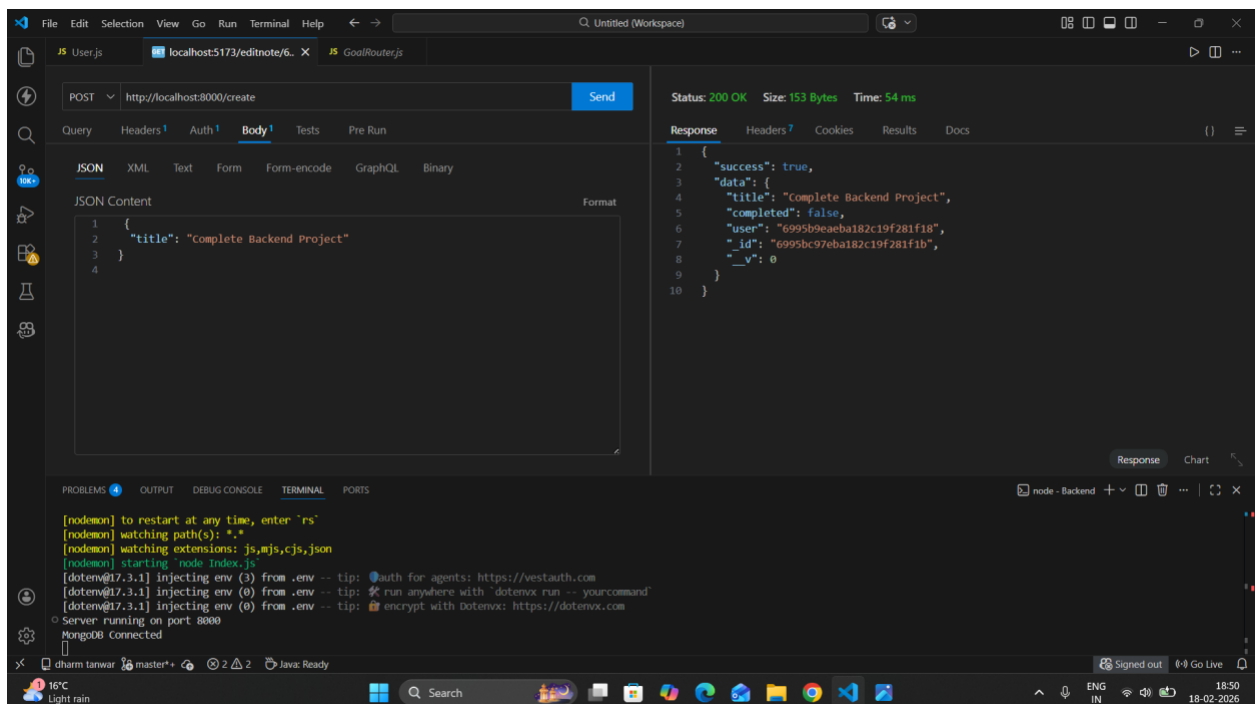


The screenshot shows a REST client interface with a status of 200 OK, size of 401 Bytes, and time of 220 ms. The response body is a JSON object with the following structure:

```
1 {
2   "success": true,
3   "message": "user is created",
4   "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQioiI20k1YjllYWVlYTE4MmMxOWYyODFmMTgiLCJpYXQiOiJE3NzE0MjAxMzgsImV4cCI6MTc3MTUwNjUzOH0.P-YJtEDk9QutEvQeMeD5HrvmBtqFVAPBz_oDiFGNqI",
5   "data": {
6     "name": "Dharam",
7     "email": "dharam123@gmail.com",
8     "password": "$2b$10$U48fBlyqlxRjOX/Q7eHieQ9B9dh2AKXhAbxArF9hs8w72Sw7zZE2",
9     "_id": "6995b9eaba182c19f281f18",
10    "_v": 0
11  }
12 }
```

The "token" field is highlighted with a blue box.

- Goal Created Successfully
- After the login and token is generated the user give post request to api for creating there goals and goal was save in MongoDB which was the database



The screenshot shows an IDE with a REST client. The request is a POST to `http://localhost:8000/create`. The request body is a JSON object:

```
1 {
2   "title": "Complete Backend Project"
3 }
```

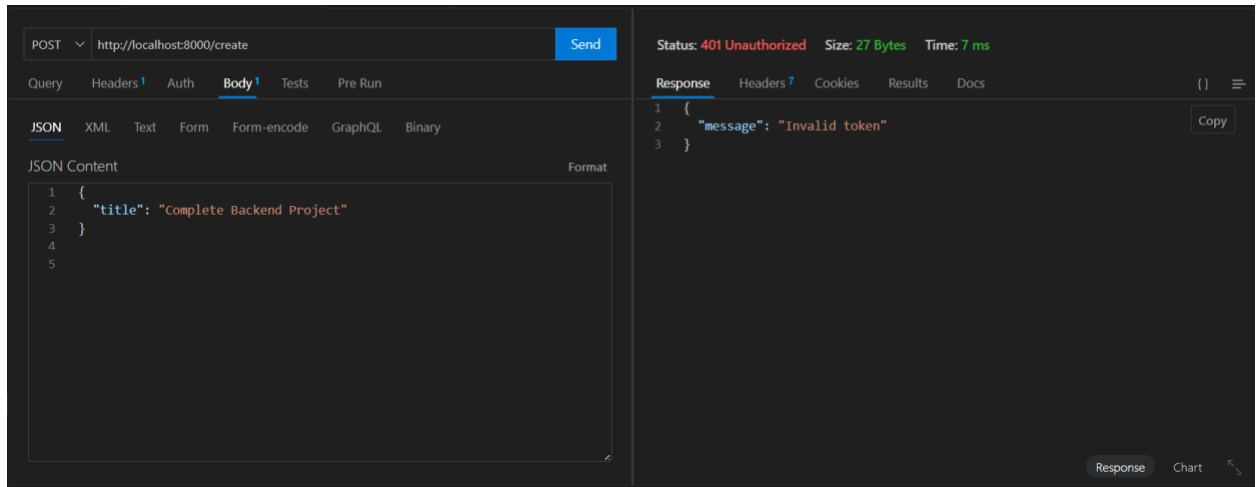
The response is a 200 OK status with 153 Bytes and 54 ms time. The response body is a JSON object:

```
1 {
2   "success": true,
3   "data": {
4     "title": "Complete Backend Project",
5     "completed": false,
6     "user": "6995b9eaba182c19f281f18",
7     "_id": "6995bc97eba182c19f281f1b",
8     "_v": 0
9   }
10 }
```

The bottom of the screenshot shows a terminal window with the following output:

```
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting node index.js
[dotenv@17.3.1] injecting env (3) from .env -- tip: auth for agents: https://vestauth.com
[dotenv@17.3.1] injecting env (0) from .env -- tip: run anywhere with `dotenvx run -- yourcommand`
[dotenv@17.3.1] injecting env (0) from .env -- tip: encrypt with Dotenvx: https://dotenvx.com
Server running on port 8000
MongoDB Connected
```

- Protected Route Working
- If the user does not have token which was generated by backend api then he cannot access the api so Route are protectedZ



Project Demo Video

Record video showing:

- Registration
- Login
- Token usage
- CRUD operations
- Error handling

Demo video : <https://drive.google.com/file/d/1nLvdAp7c8Jp7ALA8oOhbuzj9wex8ZE8c/view?usp=sharing>